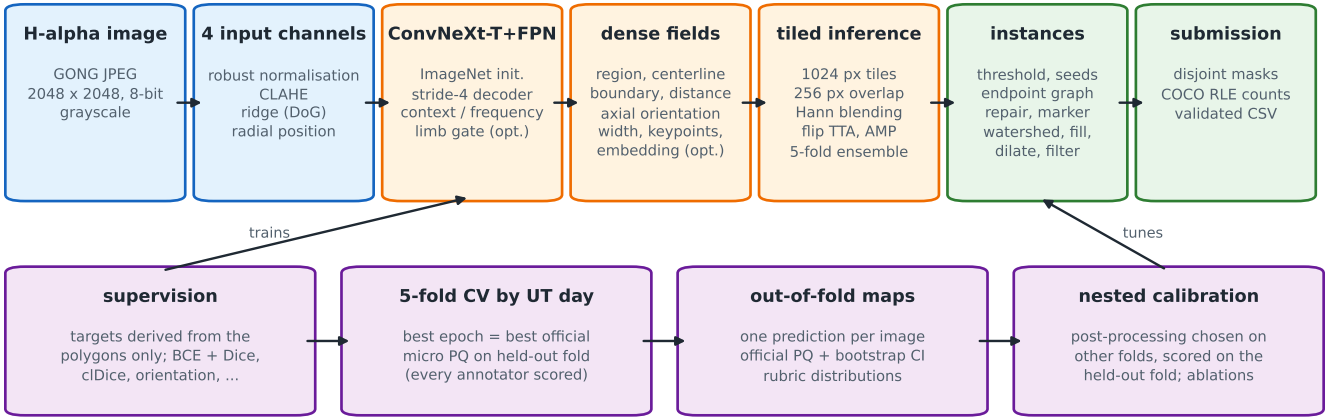
H-alpha image
GONG JPEG
2048 x 2048, 8-bit
grayscale
4 input channels
robust normalisation
CLAHE
ridge (DoG)
radial position
ConvNeXt-T+FPN
ImageNet init.
stride-4 decoder
context / frequency
limb gate (opt.)
dense fields
region, centerline
boundary, distance
axial orientation
width, keypoints,
embedding (opt.)
tiled inference
1024 px tiles
256 px overlap
Hann blending
flip TTA, AMP
5-fold ensemble
instances
threshold, seeds
endpoint graph
repair, marker
watershed, fill,
dilate, filter
submission
disjoint masks
COCO RLE counts
validated CSV
trains
tunes
supervision
targets derived from the
polygons only; BCE + Dice,
clDice, orientation, ...
5-fold CV by UT day
best epoch = best official
micro PQ on held-out fold
(every annotator scored)
out-of-fold maps
one prediction per image
official PQ + bootstrap CI
rubric distributions
nested calibration
post-processing chosen on
other folds, scored on the
held-out fold; ablations